LAB ASSIGNMENT – 7 (Studio 3t)

NAME : Pinaki Choudhuri
REGISTRATION NO. : AU/2023/0009351

ROLL NO. : UG/02/BTCSECSF/2023/012

SUBJECT : Unstructured Database Lab
SEC: E

Data: { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },

{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },

{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },

{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },

{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },

{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },

{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },

{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }.

Lab Questions:

Q1. Display all documents.

Quickstart × | students × | IntelliShell: shreya* ×

localhost:27017 > studentDB

New | Load script | Save script | Script history | Enable Query Assist | Change view | AI Helper | Visual

```
db.getCollection("students").find({})
db.students.find()
```

Raw shell output | Find Query (line 1) | Find Query (line 2)

50 | Documents 1 to 21 | Query Code | Explain Query | Table

students > roll

_id	roll	name	age	department	marks	status	grade
69c0e374955753...	1	Anu	22	IT	90	Pass	A
69c0e374955753...	2	Riya	21	CSE	78	Pass	C
69c0e374955753...	3	Aman	23	IT	92	Pass	A
69c0e374955753...	4	Neha	20	ECE	74	Pass	C
69c0e374955753...	5	Raj	24	CSE	88	Pass	B
69c0e374955753...	6	Simran	22	IT	81	Pass	B
69c0e374955753...	8	Priya	23	CSE	95	Pass	A

Q2. Find students from IT department.

Raw shell output | Find Query (line 1) | Find Query (line 2) | Find Query (line 3)

50 | Documents 1 to 17 | Query Code | Explain Query

students > roll

_id	roll	name	age	department	marks	status	grade
69c0e374955753...	1	Anu	22	IT	90	Pass	A
69c0e374955753...	3	Aman	23	IT	92	Pass	A
69c0e374955753...	6	Simran	22	IT	81	Pass	B
69c0e38f9557539...	9	Arjun	22	IT	87	Pass	B

Q3. Find students with marks > 80.

students > roll

_id	roll	name	age	department	marks	status	grade
69c0e374955753...	1	Anu	22	IT	90	Pass	A
69c0e374955753...	3	Aman	23	IT	92	Pass	A
69c0e374955753...	5	Raj	24	CSE	88	Pass	B
69c0e374955753...	6	Simran	22	IT	81	Pass	B
69c0e374955753...	8	Priya	23	CSE	95	Pass	A
69c0e38f9557539...	9	Arjun	22	IT	87	Pass	B

Q4. Insert one new student.

Result > insertedId

acknowledged	insertedId
true	69c0ef8a70a99b...

Q5. Update marks of Anu to 90.

Raw shell output | Find Query (line 1) | Find Query (line 2) | Find Query (line 3) | Find Query (line 4) | Shell Output (Documents)

50 | Documents 1 to 2

Result > insertedId

acknowledged	insertedId	matchedCount	modifiedCount	upsertedCount
true	69c0efa970a99b...			
true	null	1.0	0.0	0.0

Q6. Add a new field "status: Pass" for all students.

Raw shell output | Find Query (line 1) | Find Query (line 2) | Find Query (line 3) | Find Query (line 4) | Shell Output (Documents)

50 | Documents 1 to 3

Result > insertedId

acknowledged	insertedId	matchedCount	modifiedCount	upsertedCount
true	69c0efcc70a99b...			
true	null	1.0	0.0	0.0
true	null	24.0	3.0	0.0

Q7. Delete all students with marks < 70.

Raw shell output | Find Query (line 1) | Find Query (line 2) | Find Query (line 3) | Find Query (line 4) | Shell Output (Documents)

50 | Documents 1 to 4

Result > insertedId

acknowledged	insertedId	matchedCount	modifiedCount	upsertedCount	deletedCount
true	69c0efeb70a99b...				
true	null	1.0	0.0	0.0	
true	null	25.0	1.0	0.0	
true					0.0

Q8. Count students in CSE department.

Raw shell output | Find Query (line 1) | Find Query (line 2) | Find Query (line 3) | Find Query (line 4) | Shell Output (Documents)

50 | Documents 1 to 4

Result > insertedId

acknowledged	insertedId	matchedCount	modifiedCount	upsertedCount	deletedCount
true	69c0f02570a99b...				
true	null	1.0	0.0	0.0	
true	null	27.0	1.0	0.0	
true					0.0

Q9. Find total students in each department.

50 | Documents 1 to 3

students > total

_id	total
CSE	3.0
IT	24.0
ECE	1.0

Q10. Find highest marks.

students > roll

_id	roll	name	age	department	marks	status	grade
69c0e374955753...	8	Priya	23	CSE	95	Pass	A

Q11. Find the second highest marks.

students > roll

_id	roll	name	age	department	marks	status	grade
69c0e374955753...	3	Aman	23	IT	92	Pass	A

Q12. Find students whose name starts with 'A' and marks > 80.

students > roll

_id	roll	name	age	department	marks	status	grade
69c0e374955753...	1	Anu	22	IT	90	Pass	A
69c0e374955753...	3	Aman	23	IT	92	Pass	A
69c0e38f9557539...	9	Arjun	22	IT	87	Pass	B

Q13. Display department-wise average marks, only for departments where avg > 80.

50 | Documents 1 to 2

students > avgMarks

_id	avgMarks
CSE	87.0
IT	87.07142857142...

Q14. Find students with marks between 75 and 90, sorted by marks ascending.

students > roll

_id	roll	name	age	department	marks	status	grade
69c0e374955753...	2	Riya	21	CSE	78	Pass	C
69c0e374955753...	6	Simran	22	IT	81	Pass	B
69c0e38f9557539...	9	Arjun	22	IT	87	Pass	B

Q15. Add grade field based on marks.

50 | Documents 1 to 1

Result > insertedId

acknowledged	insertedId	matchedCount	modifiedCount	upsertedCount
true	null	34.0	13.0	0.0

```
studentDB> db.students.updateMany({}, [
...   {
...     $set: {
...       grade: {
...         $switch: {
...           branches: [
...             { case: { $gte: ["$marks", 90] }, then: "A" },
...             { case: { $gte: ["$marks", 80] }, then: "B" },
...             { case: { $gte: ["$marks", 70] }, then: "C" }
...           ],
...           default: "D"
...         }
...       }
...     }
...   }
... ])
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 8,
  upsertedCount: 0
}
```

Queries:

- db.students.find()
- db.students.find({ department: "IT" })
- db.students.find({ marks: { $gt: 80 } })
- db.students.insertOne({ roll: 9, name: "Arjun", age: 22, department: "IT", marks: 87 })
- db.students.updateOne(

{ name: "Anu" },

{ $set: { marks: 90 } }

)

- db.students.updateMany({}, { $set: { status: "Pass" } })
- db.students.deleteMany({ marks: { $lt: 70 } })

- db.students.countDocuments({ department: "CSE" })
- db.students.aggregate([

```
  { $group: { _id: "$department", total: { $sum: 1 } } }
])
```

- db.students.find().sort({ marks: -1 }).limit(1)
- db.students.find().sort({ marks: -1 }).skip(1).limit(1)
- db.students.find({

```
  name: { $regex: "^A" },
  marks: { $gt: 80 }
})
```

- db.students.find({

```
  marks: { $gte: 75, $lte: 90 }
}).sort({ marks: 1 })
```

- db.students.updateMany({}, [

```
  {
    $set: {
      grade: {
        $switch: {
          branches: [
            { case: { $gte: ["$marks", 90] }, then: "A" },
            { case: { $gte: ["$marks", 80] }, then: "B" },
            { case: { $gte: ["$marks", 70] }, then: "C" }
          ],
```

```
          default: "D"
        }
      }
    }
  }
])
```